Web Application Testing – Demoblaze (Manual & API Testing)

Name: W.A.D. Chaduki Amashi
Degree: Bachelor of Software Engineering (Hons)
University: The Open University of Sri Lanka
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Introduction

This project demonstrates both manual functional testing and REST API testing skills. Manual testing was performed on the Demoblaze e-commerce web application, while API testing was conducted using the free JSONPlaceholder service. The main objective was to verify functionality, usability, and reliability by designing and executing test cases, identifying defects, and documenting results. This hands-on project helped me understand real-world QA processes, test case design, bug reporting, and API validation using Postman.

Objectives

- To perform manual testing on a web application
- To design and execute test cases
- To identify and report bugs
- To understand real-world QA processes
- To gain experience in API testing using Postman

Tools & Technologies Used

- Browser: Google Chrome
- Testing Type: Manual Testing
- API Testing Tool: Postman
- API Source: JSONPlaceholder
- Documentation Tool: Microsoft Excel
- Platform: Demoblaze Web Application

Applications Selected for Testing

For manual functional testing, Demoblaze was chosen because it provides realistic e-commerce flows suitable for testing login, contact form, add to cart, categories filtering, and pagination modules.

For API testing practice, JSONPlaceholder was selected as a reliable, free REST API. It allowed me to practice CRUD operations, validate HTTP status codes, and verify JSON response structure without needing access to a real backend server.

Modules Tested

The following modules were tested:

- Login Module
- Contact Us Form
- Add to Cart Functionality
- Categories
- Pagination (Next/Previous Navigation)

Test Case Design

Test cases were designed based on different scenarios for each module. Each test case includes test steps, test data, expected results, and actual results. A total of 15 test cases were created and executed across five modules.

- Positive scenarios (valid inputs)
- Negative scenarios (invalid inputs)
- Edge cases (empty fields, repeated actions)

Project Name:	**Demoblaze Website Testing (Manual & API Testing)**	Total Test Cases:	15
Modules:	**Login/ Contact Us/Add to cart/ Categories/ Paginat**	Passed:	12
Test Designed By:	**W.A.D.C.Amashi**	Failed:	3
Test Designed Date:	**23/03/2026**		

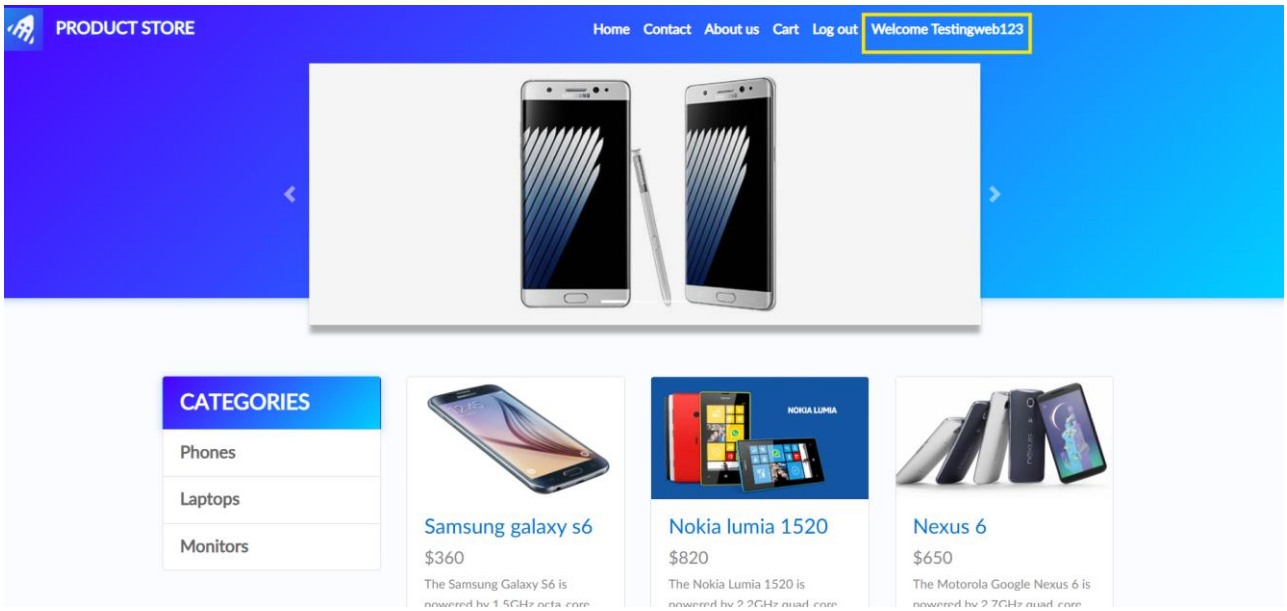
Test Scenario ID	Test Scenario	Test Case ID	Test Case Description	Pre-requisites	Test Steps	Test Data	Expected Result	Actual Result	Test	Severity
LOGIN										
TS01	Login Functionality	TC01	Verify login with valid credentials	User must be registered	1. Open login page 2. Enter Email & Password 3. Click Login	Email: Testingweb123@gmail.com Password: 12345	User logs in successfully	User logs in successfully	Pass	Low
TS02	Login Functionality	TC02	Verify login with invalid password	User must be registered	1. Open login 2. Enter Email & invalid Password 3. Click Login	Email: Testingweb123@gmail.com Password: 12346	Error message shown	Error message shown	Pass	Low
TS03	Login Functionality	TC03	Verify login with empty fields	Login page open	1. Open login 2. Click Login	None	Validation message shown	Validation message shown	Pass	Low
Contact Us										
TS04	Verify Contact Form Submission	TC04	Submit form with valid details after login	User must be Logged in	1. Open Contact form 2. Enter Email, Name, Message 3. Click Send Message	Email: Testingweb123@gmail.com Name: Testingweb Message: Website Testing	Message sent successfully	Message sent successfully	Pass	Low
TS05	Verify Contact Form Submission	TC05	Submit form without data after login	User must be Logged in	1. Open Contact form 2. Click Send Message	None	Validation message shown	Form submitted successfully even when required fields were empty	Fail	High
TS06	Verify Contact Form Submission	TC06	Submit form with data before login	User is not logged in	1. Open Contact form 2. Enter Email, Name, Message 3. Click Send Message	Email: Testingweb123@gmail.com Name: Testingweb Message: Website Testing	Error message shown	Form allows submission without requiring user login	Fail	High
Add to Cart										
TS07	Verify add to cart	TC07	Add item to cart	Site loaded	1. Select a Product 2. Click add to Cart	Product	Product added	Product added	Pass	Low
TS08	Verify add to cart	TC08	Verify cart updates	Item added	1. Add Item 2. Open cart	Product	Cart shows item	Cart shows item	Pass	Low
TS09	Verify add to cart	TC09	Remove item from cart	Item in Cart	1. Open Cart 2. Click delete	Product	Item removed	Item removed	Pass	Low
Categories										
TS10	Verify Categories	TC10	Verify selecting category	Homepage loaded	1. Click Categories (Phones/Laptops)	Category	Relevant products shown	Relevant products shown	Pass	Low
TS11	Verify Categories	TC11	Verify products match category	Category selected	1. Select Categories 2. Observe Products	Category	Correct items displayed	Correct items displayed	Pass	Low
TS12	Verify Categories	TC12	Click category multiple times	Homepage loaded	1. Click category repeatedly	Category	No crash / stable behavior	No crash / stable behavior	Pass	Low
Pagination										
TS13	Verify Pagination	TC13	Verify next button	Homepage loaded	1. Click "Next" button on homepage	None	Next products load	Next products load	Pass	Low
TS14	Verify Pagination	TC14	Verify navigation works	Multiple pages	1. Click "Next" to go to next page 2. Click "Previous" to go back	None	Navigation works correctly	Navigation works correctly	Pass	Low
TS15	Verify Pagination	TC15	Verify different items are displayed when navigating pages	User is on Home page with product list	1. Click "Next" to go to next page 2. Observe displayed products 3. Click "Previous" to go back	None	Previous page should display the same products as before	Pagination does not correctly load previous page items; some items from the next page remain	Fail	Medium

Test Execution Summary

Total Test Cases: 15 Passed: 12 Failed: 3

Status	Count
Pass	12
Fail	3

Figure 1: Successful Login

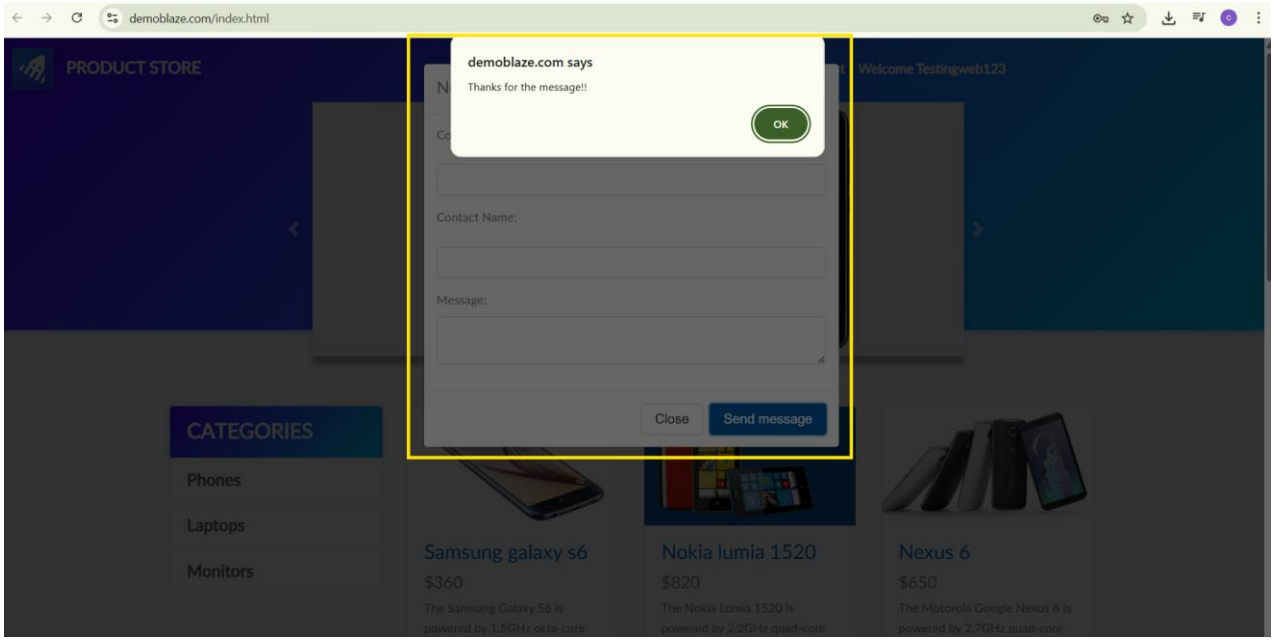


Defects (Bugs) Identified

The following defects were identified during testing:

1. Contact form allows submission even when required fields are empty (High Severity)
2. Contact form allows submission without requiring user login (High Severity)
3. Pagination does not correctly load previous page items; some items from the next page remain displayed (Medium Severity)

Figure 2: Contact form submitted without validation



API Testing

API testing was performed using JSONPlaceholder, a free online REST API for testing and prototyping. The purpose of API testing was to validate response status codes, data structure, and functionality of endpoints.

API Test Cases

API ID	Method	End Point	Description	Expected Result	Status
API01	GET	/users	Get all users	200 OK and list of users returned	Pass
API02	GET	/users/1	Get single user	200 OK and correct user data returned	Pass
API03	GET	/users/999	Invalid user	404 Not Found error returned	Pass
API04	POST	/users	Create a new user	201 Created and new user data returned	Pass
API05	PUT	/users/1	Update user	200 OK and updated data returned	Pass

Validations Performed:

- Verified HTTP status codes (200, 201, 404)
- Validated JSON response structure
- Verified correctness of returned data

Screenshots of API responses and test execution are included below:

Figure 3: Get all users

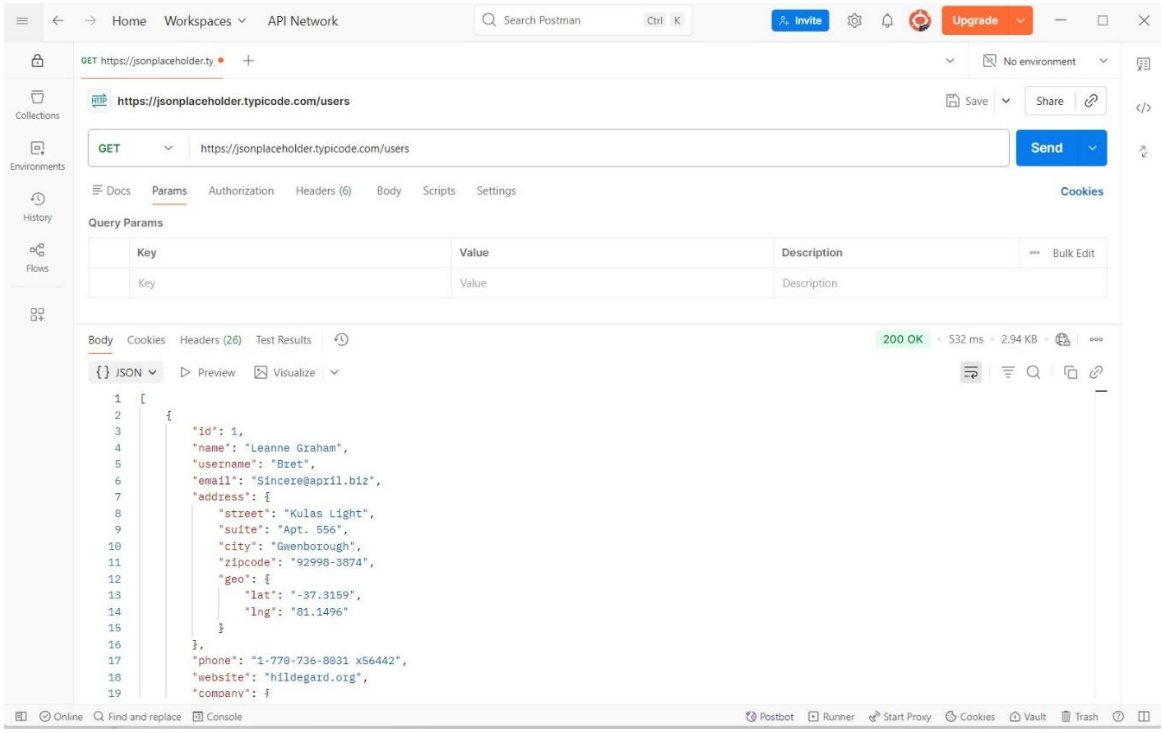


Figure 4: Get single user

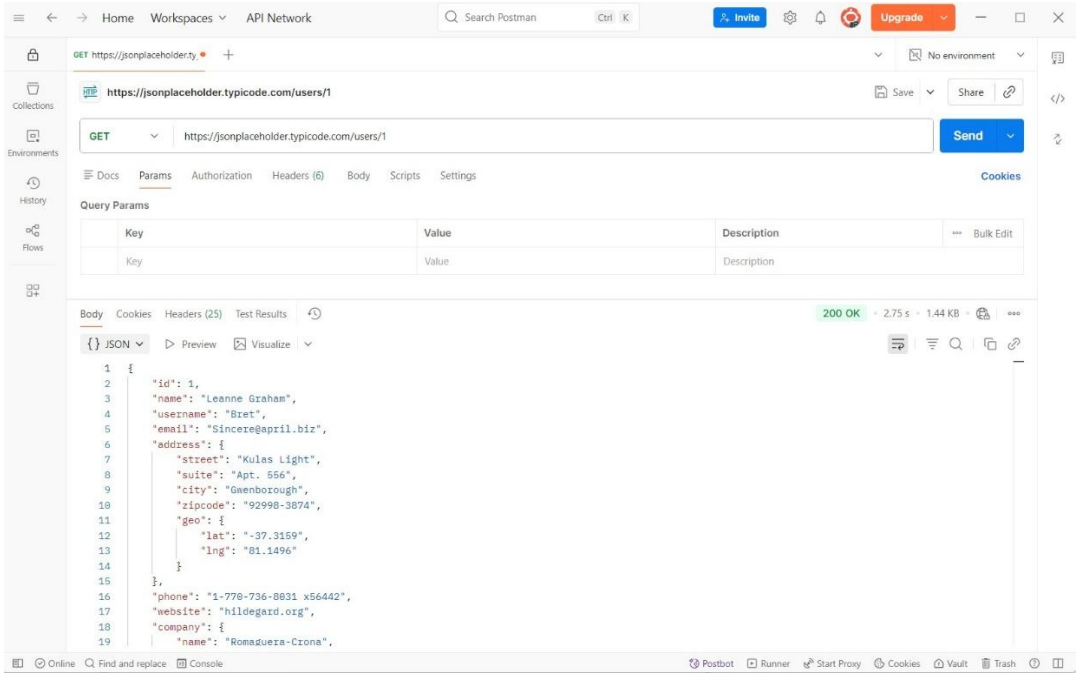


Figure 5: Invalid user

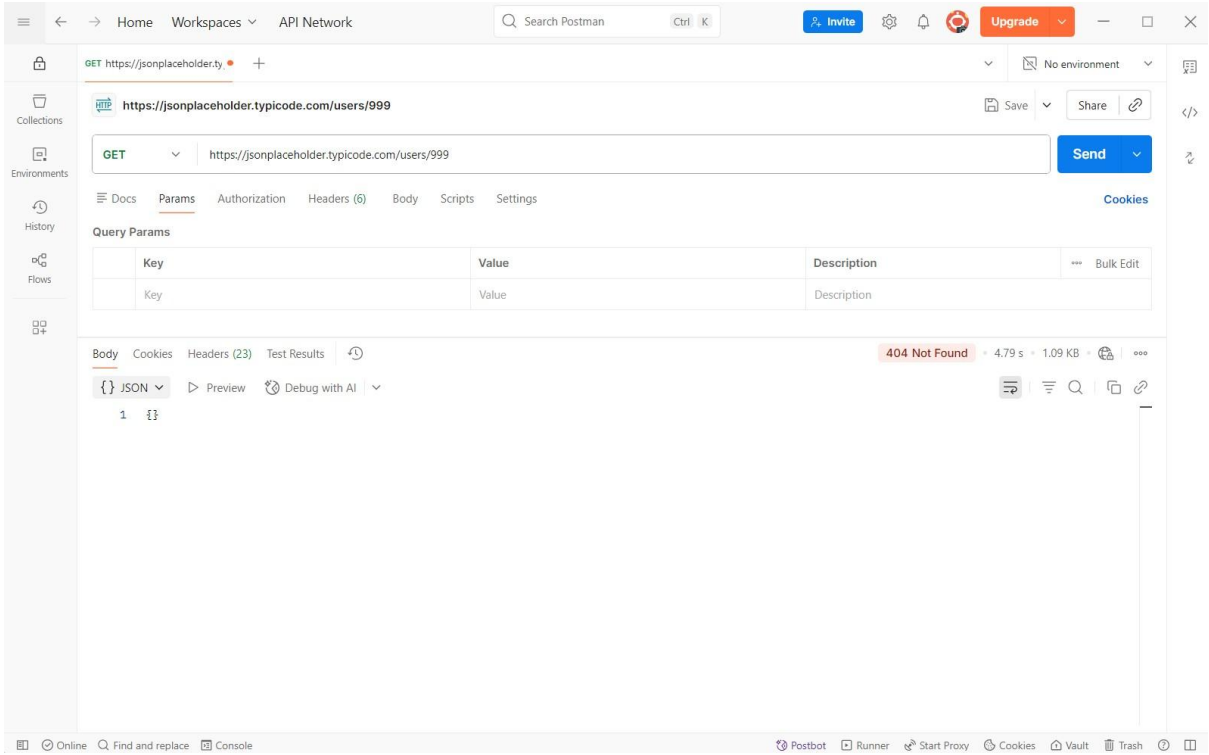


Figure 6: Create a new user

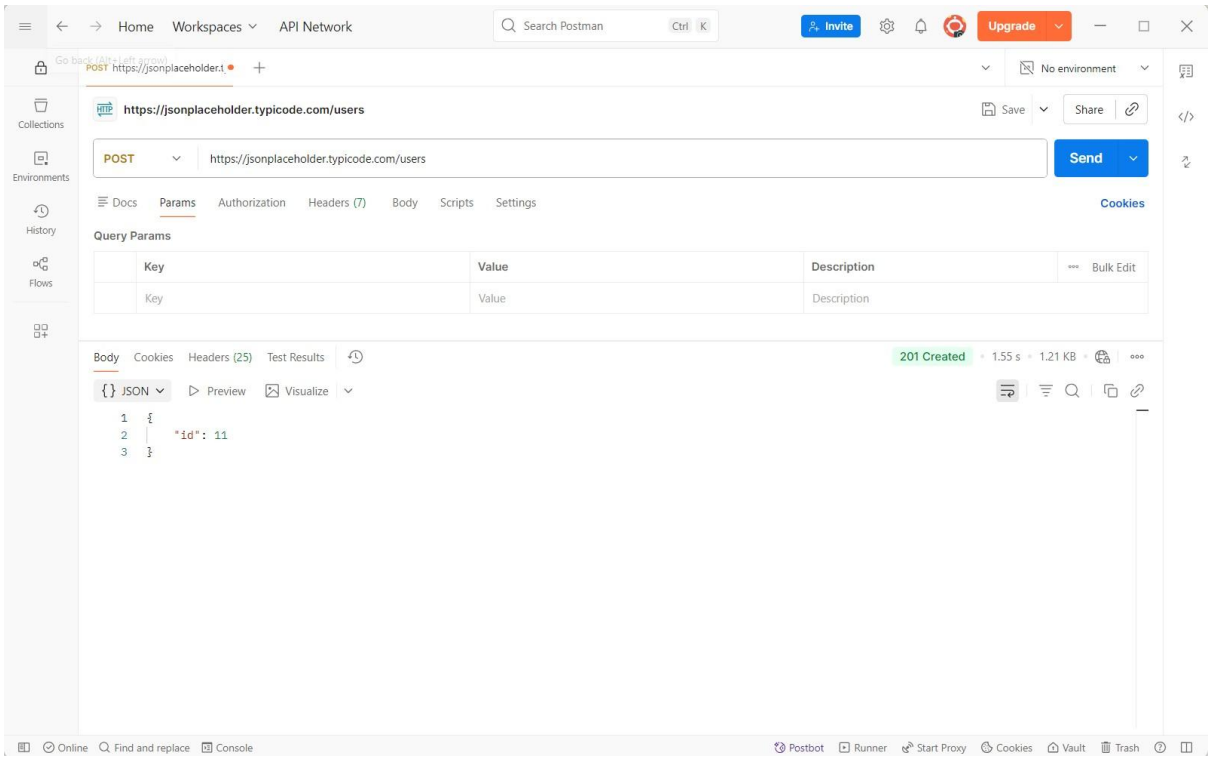
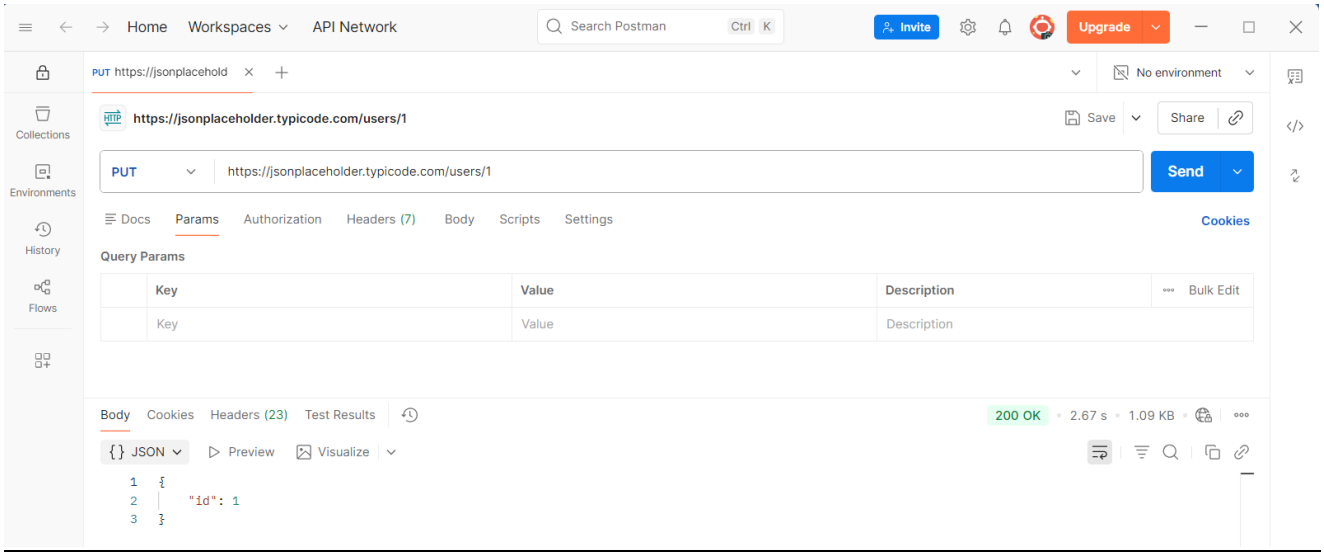


Figure 7: Update user



Challenges Faced

- Initially faced issues while testing APIs using ReqRes, then switched to JSONPlaceholder for successful testing
- Difficulty in identifying real bugs initially
- Understanding how to design proper test cases
- Learning how to use Postman for API testing

Key Learning

- Writing and executing test cases
- Identifying and reporting real defects
- Gained practical experience in real-world QA processes
- Using Postman for API validation
- Interpreting HTTP status codes

Conclusion

This project provided practical experience in manual and API testing. It helped in understanding how to design test cases, execute tests, and identify defects in a web application. The identified defects highlight potential usability and functional issues that could impact user experience if not resolved.